Han **Guo**

hanguo@mit.edu | han-guo.info | HanGuo97

Education

Massachusetts Institute of Technology — *Cambridge, MA*

PH.D. IN EECS — *Fall 2024 -*

- **Advisors:** ***Prof. Yoon Kim*** and ***Prof. Eric P. Xing***
- Language Technologies Institute, Carnegie Mellon University (2020-2024)

University of North Carolina at Chapel Hill — *Chapel Hill, NC*

COMPUTER SCIENCE WITH A SECOND MAJOR IN STATISTICS AND ANALYTICS — *2016 - 2020*

- **Advisor:** ***Prof. Mohit Bansal (UNC-NLP Lab)***

Publications

CODA: Rewriting Transformer Blocks as GEMM-Epilogue Programs [pdf]
Han Guo, *Jack Zhang, Arjun Menon, Driss Guessous, Vijay Thakkar, Yoon Kim, Tri Dao.*
NeurIPS 2026

BiLoCo: Binary Low-Rank Corrections for Post-Training Quantization [pdf]
David Jin, Beshr IslamBouli*, Tarushii Goel,* ***Han Guo****, Yoon Kim.*
NeurIPS 2026

Fast KV Compaction via Attention Matching [pdf]
Adam Zweiger, Xinghong Fu, ***Han Guo****, Yoon Kim.*
ICML 2026

Self-Adapting Language Models [pdf]
Adam Zweiger, Jyothish Pari*,* ***Han Guo****, Ekin Akyürek, Yoon Kim, Pulkit Agrawal.*
NeurIPS 2025

Log-Linear Attention [pdf]
Han Guo*, Songlin Yang*, Tarushii Goel, Eric P. Xing, Tri Dao, Yoon Kim.*
ICLR 2026

On the Duality between Gradient Transformations and Adapters [pdf]
Lucas Torroba-Hennigen, Hunter Lang, ***Han Guo****, Yoon Kim.*
ICML 2025

Pushing the Limits of Large Language Model Quantization via the Linearity Theorem [pdf]
Vladimir Malinovskii, Andrei Panferov*, Ivan Ilin,* ***Han Guo****, Peter Richtárik, Dan Alistarh.*
NAACL 2025

The Surprising Effectiveness of Test-Time Training for Few-Shot Learning [pdf]
Ekin Akyürek, Mehul Damani, Adam Zweiger*, Linlu Qiu,* ***Han Guo****, Jyothish Pari, Yoon Kim, Jacob Andreas.*
ICML 2025

Training-Free Activation Sparsity in Large Language Models [pdf]
James Liu, Pragaash Ponnusamy, Tianle Cai, ***Han Guo****, Yoon Kim, Ben Athiwaratkun.*
ICLR 2025 (Spotlight)

Fast Matrix Multiplications for Lookup Table-Quantized LLMs [pdf]
Han Guo*, William Brandon, Radostin Cholakov, Jonathan Ragan-Kelley, Eric P. Xing, Yoon Kim.*
EMNLP 2024 Findings

LQ-LoRA: Low-rank Plus Quantized Matrix Decomposition for Efficient Language Model Finetuning [pdf]
Han Guo*, Philip Greengard, Eric P. Xing, Yoon Kim.*
ICLR 2024

Federated Learning as Variational Inference: A Scalable Expectation Propagation Approach [pdf]
Han Guo*, Philip Greengard, Hongyi Wang, Andrew Gelman, Yoon Kim*, Eric P. Xing*.*
ICLR 2023

MPCFormer: Fast, Performant and Private Transformer Inference with MPC [pdf]
Dacheng Li, Rulin Shao*, Hongyi Wang*, **Han Guo**, Eric P. Xing, Hao Zhang.*
ICLR 2023 (Notable-top-25%)

RLPrompt: Optimizing Discrete Text Prompts With Reinforcement Learning [pdf]
Mingkai Deng, Jianyu Wang*, Cheng-Ping Hsieh*, Yihan Wang, **Han Guo**, Tianmin Shu, Meng Song, Eric P. Xing, Zhiting Hu.*
EMNLP 2022

Efficient (Soft) Q-Learning for Text Generation with Limited Good Data [pdf]
***Han Guo**, Bowen Tan, Zhengzhong Liu, Eric P Xing, Zhiting Hu.*
EMNLP 2022 Findings

VisualGPT: Data-efficient Image Captioning by Balancing Visual Input and Linguistic Knowledge from Pretraining [pdf]
*Jun Chen, **Han Guo**, Kai Yi, Boyang Li, Mohamed Elhoseiny.*
CVPR 2022

Uncertainty Toolbox: an Open-Source Library for Assessing, Visualizing, and Improving Uncertainty Quantification [pdf]
*Youngseog Chung, Ian Char, **Han Guo**, Jeff Schneider, Willie Neiswanger.*
Uncertainty and Robustness in Deep Learning Workshop, ICML 2021

FastIF: Scalable Influence Functions for Efficient Model Interpretation and Debugging [pdf]
***Han Guo**, Nazneen Fatema Rajani, Peter Hase, Mohit Bansal, Caiming Xiong.*
***EMNLP 2021**.*

An Overview of Uncertainty Calibration for Text Classification and the Role of Distillation [pdf]
***Han Guo**, Ramakanth Pasunuru, and Mohit Bansal.*
RepL4NLP Workshop, ACL 2021

DORB: Dynamically Optimizing Multiple Rewards with Bandits [pdf]
*Ramakanth Pasunuru, **Han Guo**, and Mohit Bansal.*
***EMNLP 2020**.*

Multi-Source Domain Adaptation for Text Classification via DistanceNet-Bandits [pdf]
***Han Guo**, Ramakanth Pasunuru, and Mohit Bansal.*
AAAI 2020

AutoSeM: Automatic Task Selection and Mixing in Multi-Task Learning [slides] [pdf]
***Han Guo**, Ramakanth Pasunuru, and Mohit Bansal.*
NAACL 2019

Dynamic Multi-Level Multi-Task Learning for Sentence Simplification [pdf]
***Han Guo**, Ramakanth Pasunuru, and Mohit Bansal.*
***COLING 2018** (“Area Chair Favorites” Paper Award)*

Soft, Layer-Specific Multi-Task Summarization with Entailment and Question Generation [pdf]
Han Guo, Ramakanth Pasunuru*, and Mohit Bansal.*
ACL 2018

Interactive-Length Multi-Task Video Captioning with Cooperative Feedback [link]
***Han Guo**, Ramakanth Pasunuru, and Mohit Bansal.*
NeurIPS 2017 (demo papers)

Towards Improving Abstractive Summarization via Entailment Generation [pdf]
*Ramakanth Pasunuru, **Han Guo**, and Mohit Bansal.*
Workshop on Summarization Frontiers, EMNLP 2017

Experience

Research Intern — *San Francisco, CA*
Together AI — *2025.6 - 2026.2*
- Advisor: Tri Dao

Research Intern — *Cambridge, MA*
MIT-IBM Watson AI Lab — *2021.6 - 2021.8*
- Advisor: Yoon Kim

Research Intern *Palo Alto, CA*

Salesforce Research *2020.6 - 2020.8*

- Advisors: Caiming Xiong, Nazneen Fatema Rajani

Research Intern *Sunnyvale, CA*

Baidu Research *2019.5 - 2019.8*

- Advisor: Boyang Li

Awards/Honors

2026.04	**Jane Street Graduate Research Fellowship**
2024.12	**2nd place in ARC Prize (Paper Award)**
2023.08	**Machine Learning and Systems Rising Stars**
2022.10	**Microsoft Research Ph.D. Fellowship**
2019.12	**Finalist, 2020 CRA Outstanding Undergraduate Researchers**
2018.6	**Area Chair Favorites Paper Award, COLING 2018**